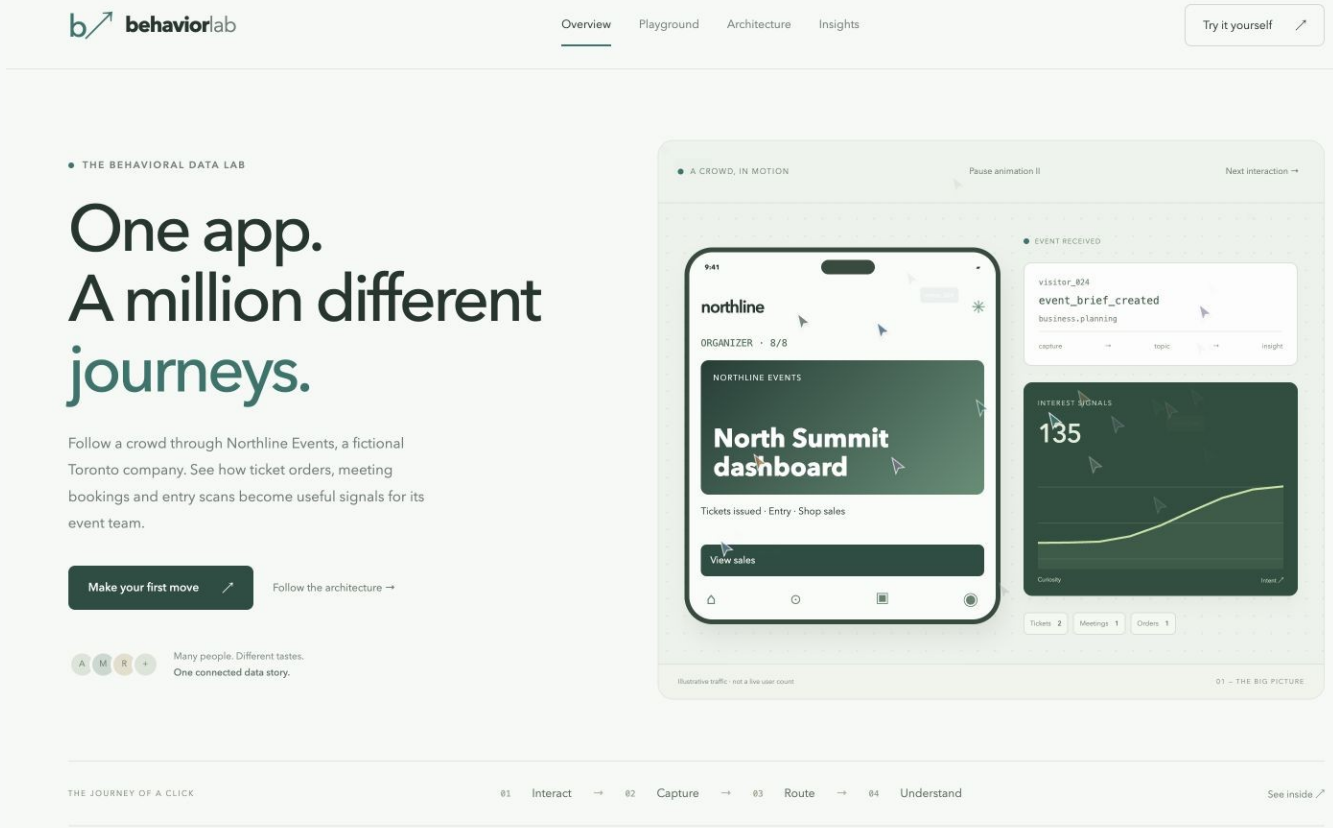
Behavior Lab

Behavioral analytics for a corporate-events application

Northline Events is a fictional Canadian company.
A working phone experience makes event capture and client insights inspectable.



The event-business problem

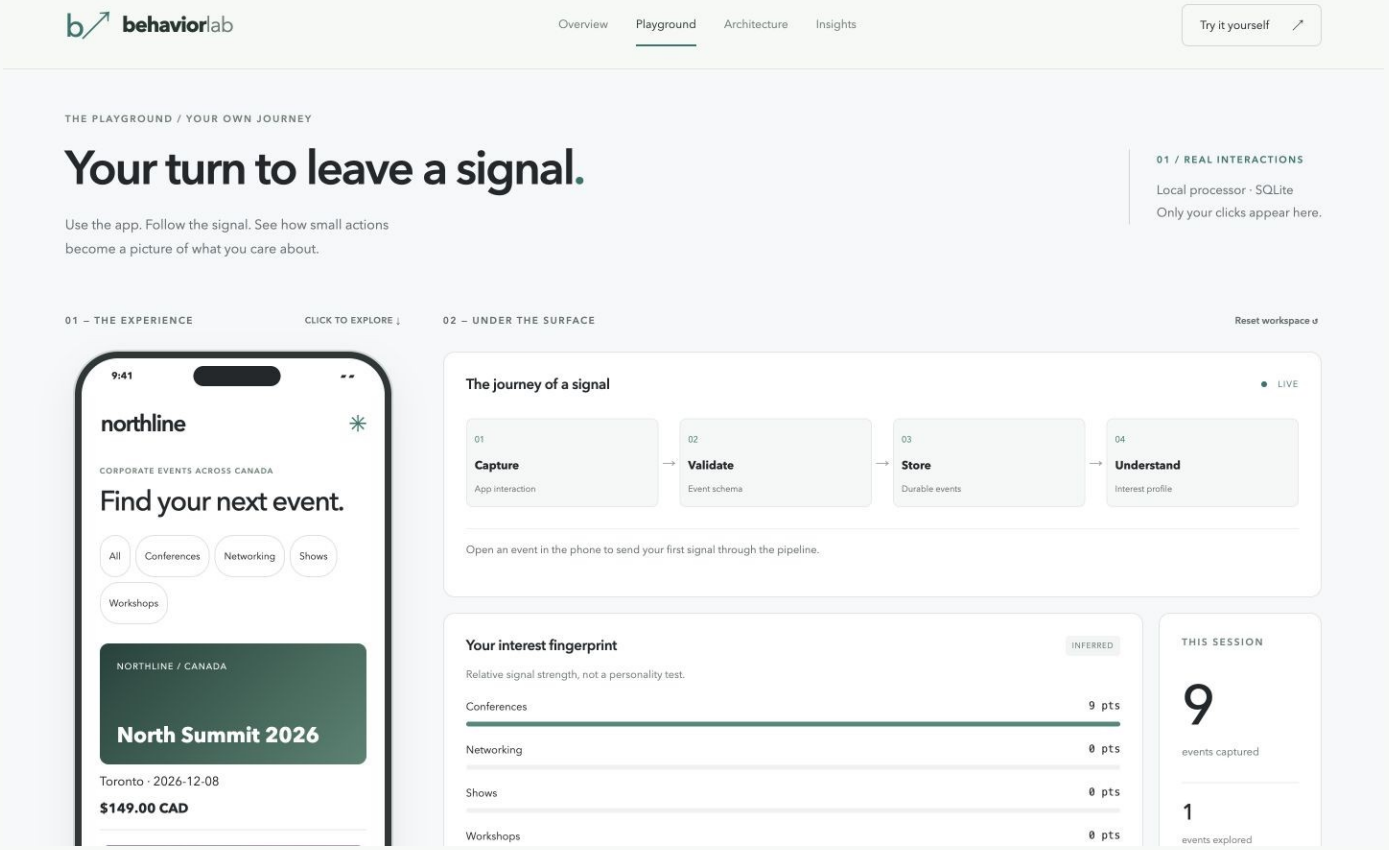
Connect client activity with commercial follow-up

Event browsing, ticket orders and meeting requests often live in separate systems. Behavior Lab records the action, preserves its source and explains the resulting profile.

A complete demo journey

Browse, buy, plan and attend

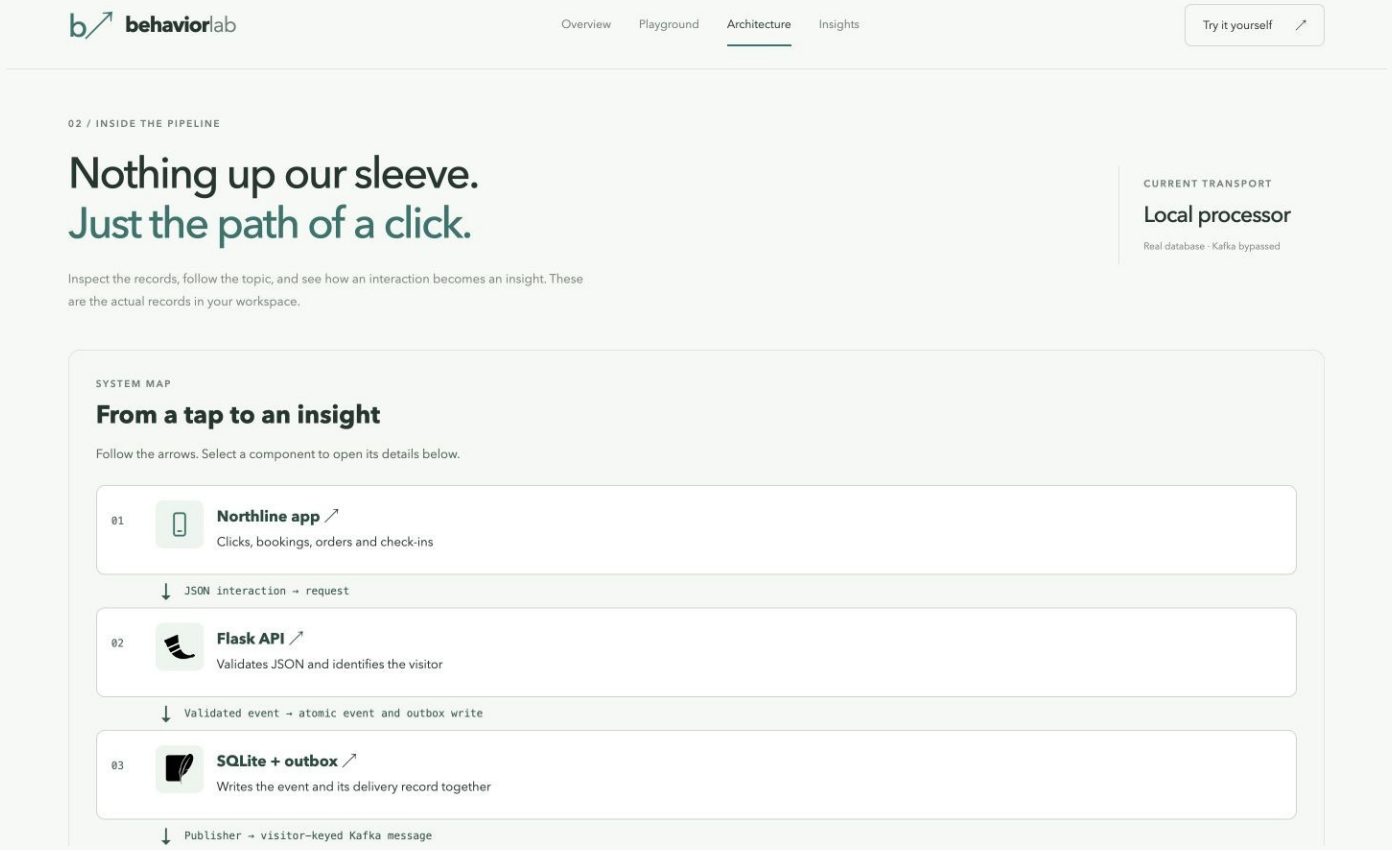
Discover Canadian events and order tickets or merchandise.
Book a consultation, prepare a brief and request a proposal.
Issue tickets, simulate entry and reconcile attendance with sales.



Event processing architecture

Capture and delivery use a transactional outbox

Flask API → SQLite event and outbox → Kafka topics → Python consumer. The consumer commits its offset after database materialization succeeds.



Data layers and recovery

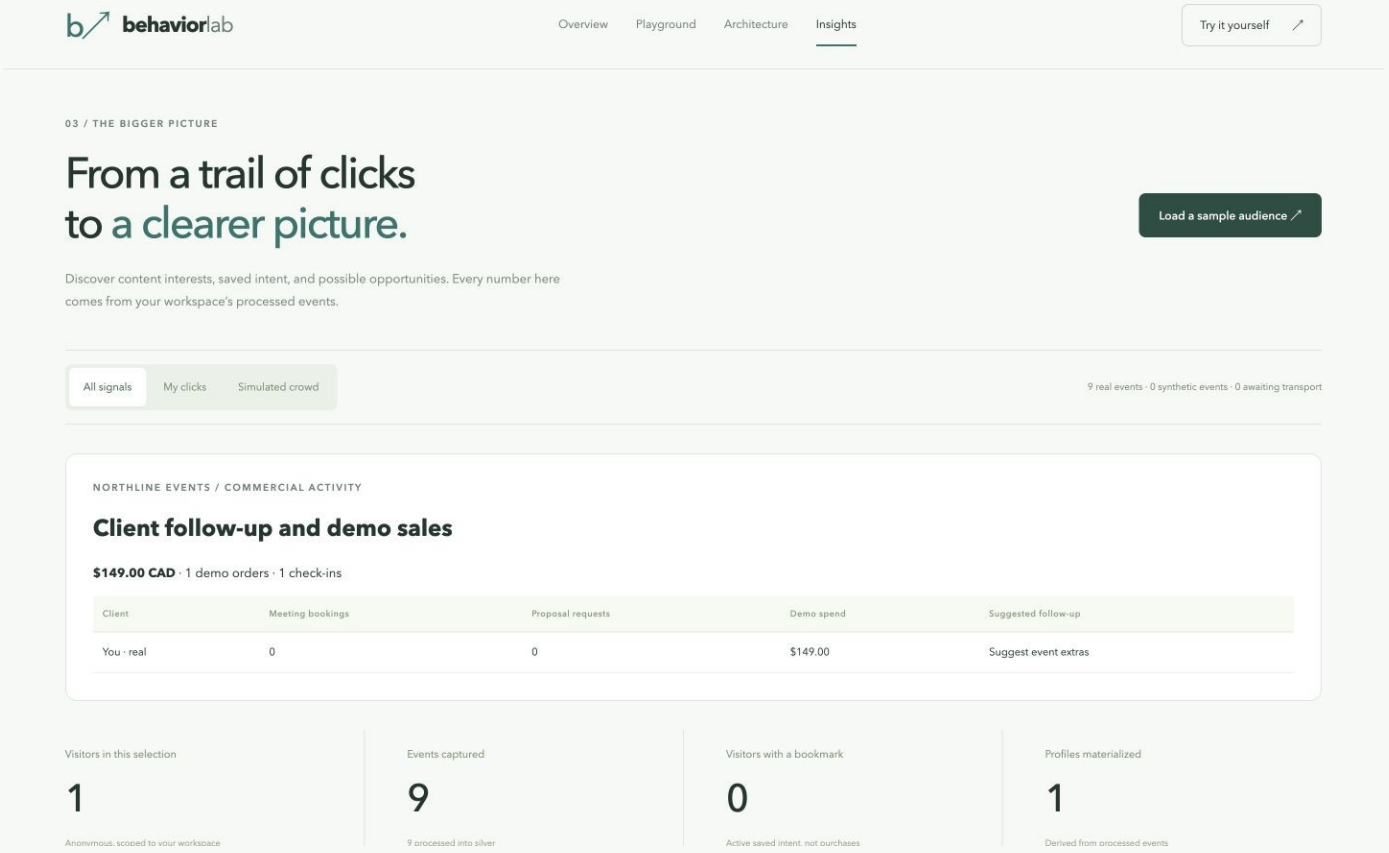
Raw events support consistent profile reconstruction

Bronze retains raw payloads. Silver records effective contributions.
Gold stores visitor preferences and active saved items.
Unique event IDs support duplicate delivery and capture-sequence replay.

Client evidence and sales

Follow-up suggestions use explicit activity

Meetings and proposal requests identify conversations to prepare for.
Ticket and merchandise orders contribute demo revenue.
Source filters distinguish the synthetic cohort from real demo interactions.



Validation and current scope

Local behavior is verified

17 pipeline tests passed. API checks reconciled CAD 330 in demo purchases.
Browser checks reconciled a ticket order and check-in across the app and reports.
GitHub CI passed a real Kafka smoke test on 5 November 2026.
Public backend hosting remains pending.

Repository and next steps

Run, inspect and extend the project

github.com/adityamalviya04692/Project-BehaviorLab
Python / Flask / SQLite / Kafka integration / Docker / GitHub Actions
Prepared for a Docker deployment with a real-broker smoke test.